Binomial random variables

1

a Independent

b Two possible outcomes: success and failure.

2

a Yes **b** No **c** Yes **d** Yes

e Yes **f** No

3

a The probability of success. **b** The probability of failure.

4

a 0.3 **b** 6

c The probability of exactly 2 successes.

5

a $\binom{4}{0}p^0q^4 + \binom{4}{1}p^1q^3 + \binom{4}{2}p^2q^2 + \binom{4}{3}p^3q^1 + \binom{4}{4}p^4q^0$

b The probability of exactly 2 successes.

Distributions of binomial random variables

6

a $p = 0.5$ **b** $p = 0.15$ **c** $p = 0.85$

7

a Equal **b** Symmetric

8

a Greater than **b** Positively skewed

9

a Less than **b** Negatively skewed

10

a

i Symmetric **ii** $p = 0.50$

b

i Positively skewed **ii** $p = 0.21$

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c

i Negatively skewed ii $p = 0.80$

Probability for binomial random variables

11

a 0.00046 **b** 0.94208

12

a $P(X = 2) = \binom{13}{2} \times (0.4)^2 \times (0.6)^{11}$ **b** 0.0453

13

a 0.0425 **b** $n = 10$ **c** $p = 0.6$

14

a $n = 4$ **b** $p = 0.4$

c ${}^4C_2 \times (0.4)^2 \times (0.6)^{4-2}$ **d** 6

e $P(X = 2) = 0.3456$

15

x	0	1	2	3
$P(X = x)$	0.343	0.441	0.189	0.027

16

a ${}^4C_1 \times (0.1)^1 \times (0.9)^{4-1}$ **b** ${}^4C_2 \times (0.1)^2 \times (0.9)^{4-2}$

c

x	0	1	2	3	4
$P(X = x)$	0.6561	0.2916	0.0486	0.0036	0.0001

17

a $n = 20$ **b** $p = 0.75$

c $P(X = 11) = 0.0271$

18

a $n = 6$ **b** $1 - p = 0.2$ **c** $\frac{256}{3125}$ **d** $\frac{309}{3125}$

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a $\frac{992}{3125}$ **b** $\frac{2133}{3125}$

20

a 5 **b** 0.2

c $P(X > 3) = \frac{21}{3125}$ **d** 1

e 1 **f** 0.8

21

a 8 **b** $\frac{4}{5}$ **c** 0.94 **d** 7

e $\frac{32}{5}$ **f** $\frac{32}{25}$

22

a 0.52822 **b** 0.16308

23

a 4 **b** 0.0756 **c** 0.9163 **d** $p = 0.3$

e 1.2 **f** $\frac{\sqrt{21}}{5}$

24

a 5 **b** 0.31744 **c** 0.68256 **d** 0.07776

e $p = 0.6$ **f** $\sqrt{\frac{6}{5}}$

25

a 4 **b** 0.0256 **c** 0.9728 **d** $p = 0.2$

e 0.8 **f** $\frac{4}{5}$

26

a 5 **b** 0.83692 **c** 0.16308 **d** 0.00243

e $p = 0.3$ **f** $\sqrt{\frac{21}{20}}$

27

a 0.07 **b** 0.28 **c** 0.75 **d** 0.41

e 0.22

28

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a $p = 0.6$ **b** 2.4 **c** 0.96

29

a $p = \frac{3}{5}$ **b** $\frac{12}{5}$

30

a $n = 40$ **b** $p = 0.2$

31

a $n = 25$ **b** $\frac{\sqrt{21}}{2}$

32

a $p = 0.2$ **b** $n = 10$

33

a $p = 0.4$ **b** $n = 12$

Applications

34 70 people

35 65

36

a $\frac{1}{6}$ **b** 0.1608 **c** 0.0322

37

a $\frac{3}{11}$ **b** 0.2861 **c** 0.2035

38

a 0.2212 **b** 0.0125

39

a $\binom{5}{2} \times \left(\frac{1}{5}\right)^2 \times \left(\frac{4}{5}\right)^3$

b The probability of selecting exactly 3 milk chocolates.

c $\frac{256}{625}$

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d $\frac{2304}{3125}$

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